

8TH GRADE MATH • SUPPORT NOTES

Unit 4: Angle Relationships

NC.8.G.5

Name: _____

Date: _____

Class: _____

Period: _____

★ The One Big Idea

Angles come in **pairs that follow rules**. If you know one angle, the rules help you find the others.

□ Words to Know

Angle — the opening between two rays, measured in degrees ($^{\circ}$)

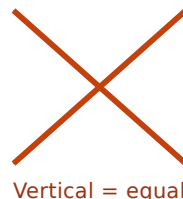
Parallel lines — two lines that never cross (same distance apart)

Transversal — a line that crosses two other lines

Congruent — equal in size

4.1 Angle Relationships

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**The Four Pairs**

Pair	Rule
Complementary	two angles add to 90°
Supplementary	two angles add to 180°
Vertical	across from each other — equal
Adjacent	next to each other, share a side

I Do

Two angles are complementary. One is 35° . The other is $90 - 35 = \mathbf{55^\circ}$.

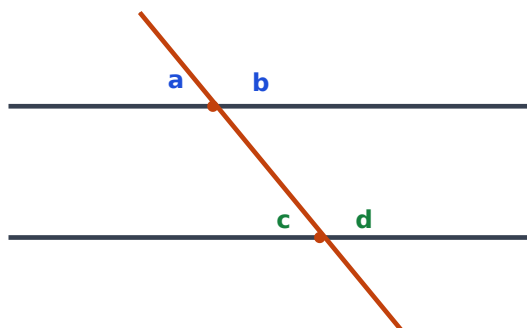
Two angles are supplementary. One is 110° . The other is $180 - 110 = \mathbf{70^\circ}$.

You Try

Complement of $62^\circ = \underline{\hspace{1cm}}^\circ$ Supplement of $145^\circ = \underline{\hspace{1cm}}^\circ$

4.2 Parallel Lines & Transversals

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A transversal cuts two parallel lines

▤ Angle Pairs That Are EQUAL

- **Corresponding** (same corner) — equal
- **Alternate interior** (inside, opposite sides) — equal
- **Alternate exterior** (outside, opposite sides) — equal
- **Vertical** — equal

★ Angle Pairs That ADD TO 180°

- **Same-side interior** (co-interior) — supplementary
- Any angle + the one next to it on a line — supplementary

Quick check: every angle is either equal to a given angle, or adds with it to 180°.

4.3 Triangle Angle Theorems

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★ Two Triangle Rules

Angle Sum: the three inside angles of any triangle add to **180°**.

Exterior Angle: an outside angle equals the **two far inside angles added together**.

👤 I Do

A triangle has angles 50° , 60° , and x . So $50 + 60 + x = 180 \rightarrow x = 70^\circ$.

Exterior angle = $50^\circ + 60^\circ = 110^\circ$.

✎ You Try

A triangle has 90° and 35° . The third angle = _____°

Part 2 Practice **Find each angle**

#	Question	Answer
1	Complement of 28°	
2	Supplement of 73°	
3	Vertical angle to 115°	
4	Triangle: 45° , 55° , third = ?	
5	Two parallel lines cut by a transversal; one angle is 70° . Its corresponding angle = ?	
6	Exterior angle when remote angles are 40° and 65°	

 Sentence Frame

"These angles are ____, so they ____ (are equal / add to 90° / add to 180°)."

★ ANSWER KEY ★ **Guided Notes**

4.1: $28 \rightarrow$ complement 62° ? (62°); supplement $145 \rightarrow 35^\circ$. You try: 28° , 35° .

4.3: $90 + 35 \rightarrow$ third = 55° .

 Practice

1) 62° 2) 107° 3) 115° 4) 80° 5) 70° 6) 105°

End of Unit 4 support packet.